ICSIE 2026 · FULL PAPER PRESENTATION

SportSpace: Spatial Evidence for Planning Active Urban Environments in Ilidža–Stup

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Interactive pilot & paper: 3esign.github.io/sportspace-pilot

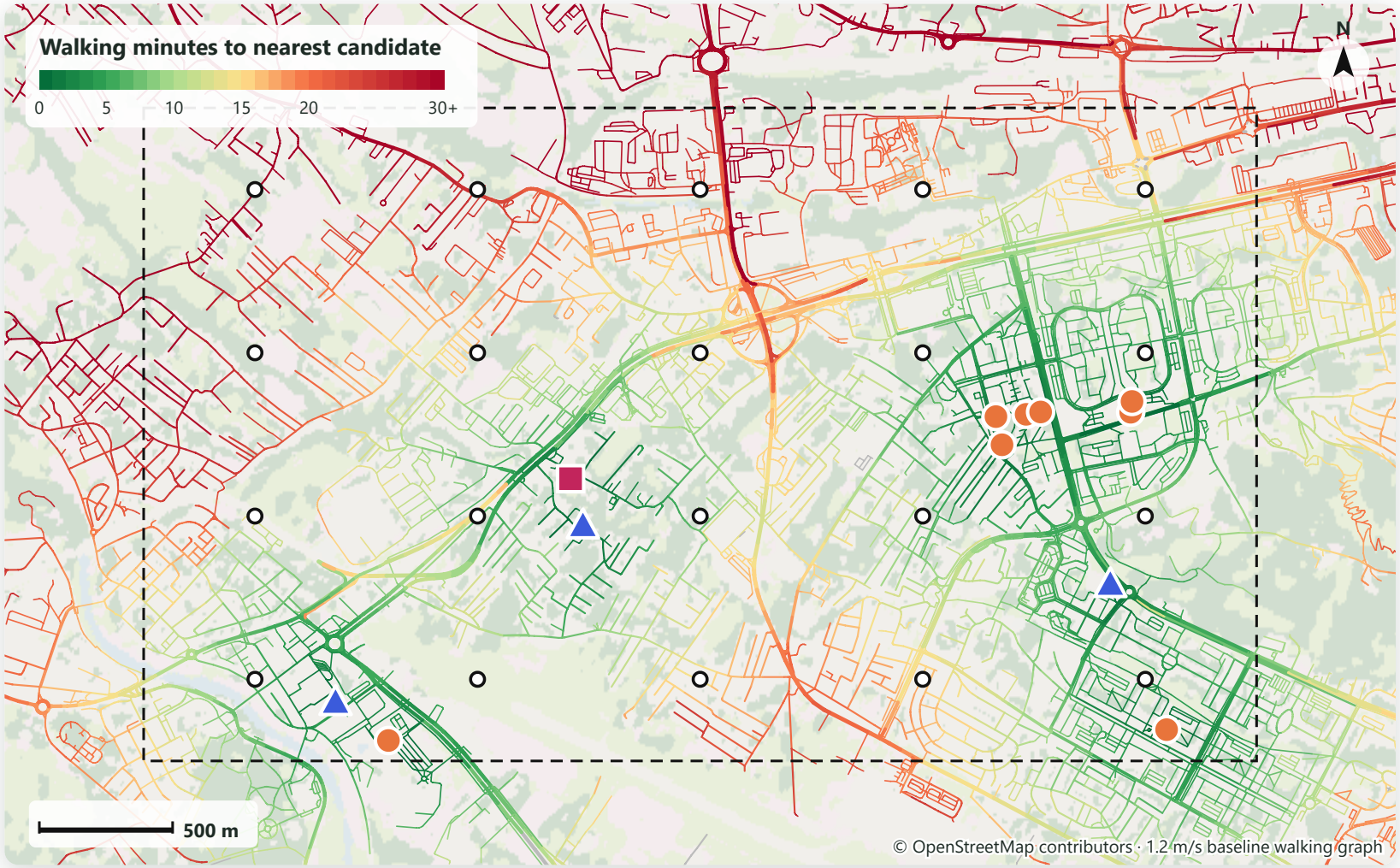


A map symbol is not an opportunity

The same symbols sit inside very different walking geographies.

1. Mapped candidate ≠ usable opportunity
2. Modelled opportunity ≠ actual participation
3. Health-relevant conditions ≠ demonstrated health effect

WHO (2018, 2020): active environments belong to health-oriented planning. The early question: which candidates merit a closer look, what does reaching them cost, and what could change the answer?



Every street segment coloured by modelled walking minutes to the nearest of the 12 candidates (21,327 nodes, 1.2 m/s).

Question, contribution and position

> *How can modelled access, green context and historical environmental conditions be translated into transparent planning questions for sport and recreation, without treating them as verified participation or health outcomes?*

Proximity

Straight-line vs. network distance, per activity category

Green scale

Sensitivity of travel-time / vegetation trade-offs to buffer radius

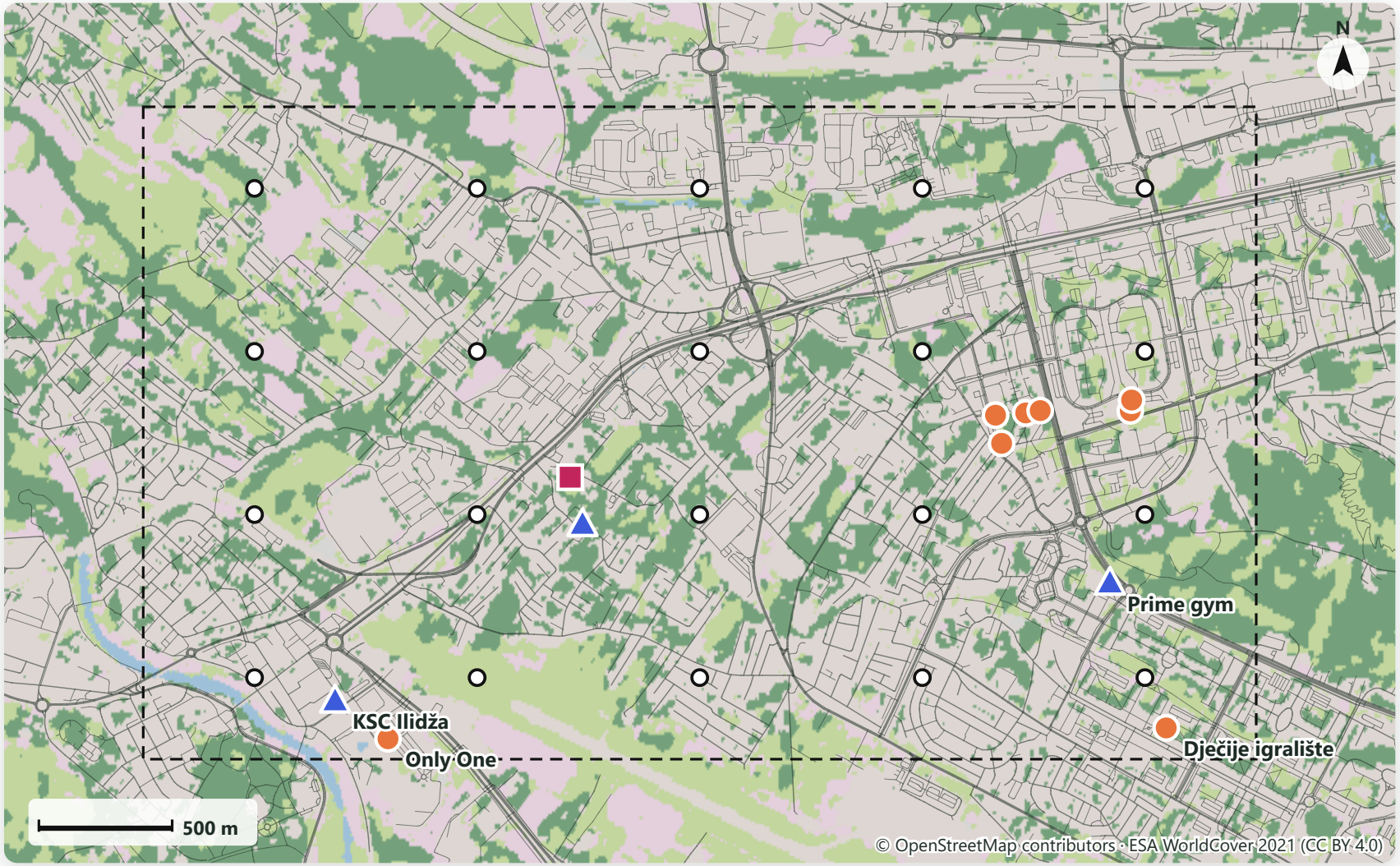
Timing

Pollutant-specific comparison of historical departure times

Strand	Taken from it	Not claimed
Health-oriented planning (WHO 2016–2020)	Motivation to examine activity-supporting environments	Conversion from minutes or vegetation % to health effect
Accessibility measures (Geurs & van Wee 2004)	Nearest, cumulative, decay-weighted as complementary	Calibrated local preferences
Pollution-aware routing (Yan 2024) · Green & air (Venter 2024) · Vis design (Munzner 2009)	Trade-offs in travel; vegetation separate from air; interface as instrument	Street-level routing; vegetation as pollution reducer; proof the interface improves decisions

Contribution: an applied, reproducible workflow connecting each indicator to a decision and to the evidence required before acting on it.

Study design and data: Ilidža–Stup pilot



ESA WorldCover 2021 land cover · OSM 2026 walking graph (4,407 ways) · technical window 18.300–18.352° E, 43.828–43.850° N

20
synthetic origins · 19 routable

12
mapped candidates

228
of 240 OD pairs routable

8,760
hours of 2021 air & weather

- Playground (8)
- Sports centre (3)
- Pitch (1)
- Synthetic origin (20)
- Technical window

Walking 1.2 m/s · 350 m snapping · oneway directed. A 2026 network against 2021 environment is a declared scenario, not one observed year.

Method: three measures, one trade-off, one integral

Accessibility

Nearest destination: straight-line vs. network

Cumulative: $N_i(\tau) = \Sigma_j 1[t_{ij} \leq \tau]$, τ = 10 / 20 / 30 min

Potential: $P_i(h) = \Sigma_j 2^{-t_{ij}/h}$, h = 10 / 15 / 30 min

Green trade-off

$V_j(r)$: vegetation % at r = 100 / 300 / 500 m

Joint count with threshold g = 10 / 20 / 30 %

Pareto set: minimise t_{ij}, maximise $V_j(r)$

1,620 parameter rows; 57 radius comparisons per activity

Outing scenarios

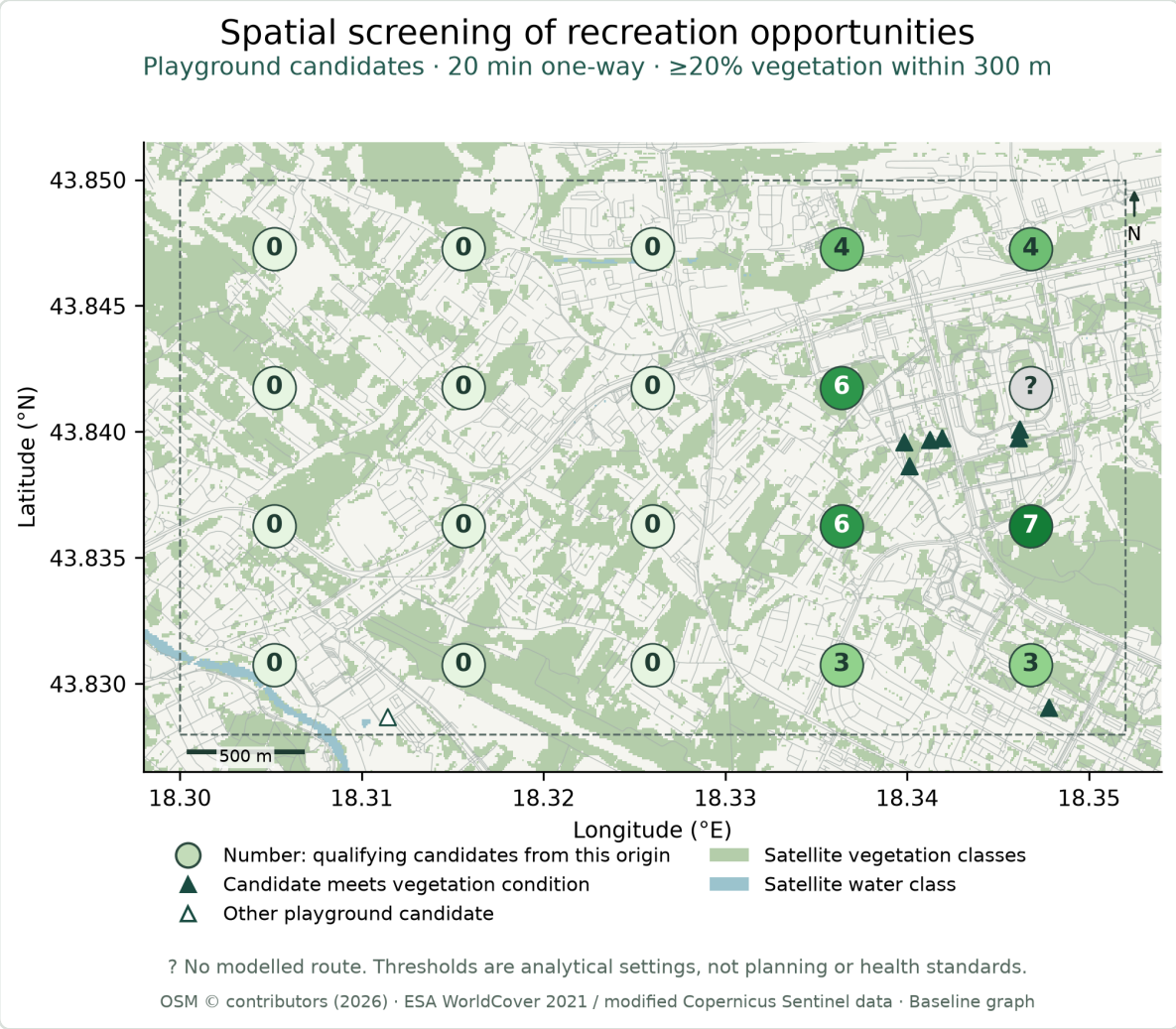
Departures 08:00 / 14:00 / 18:00, every 2021 date; stay 30 min (15, 60)

$E_p = \int_0^T C_p(s+u)\, du$, $T = 2t + d$

83,220 paired cases per pollutant; not inhaled dose

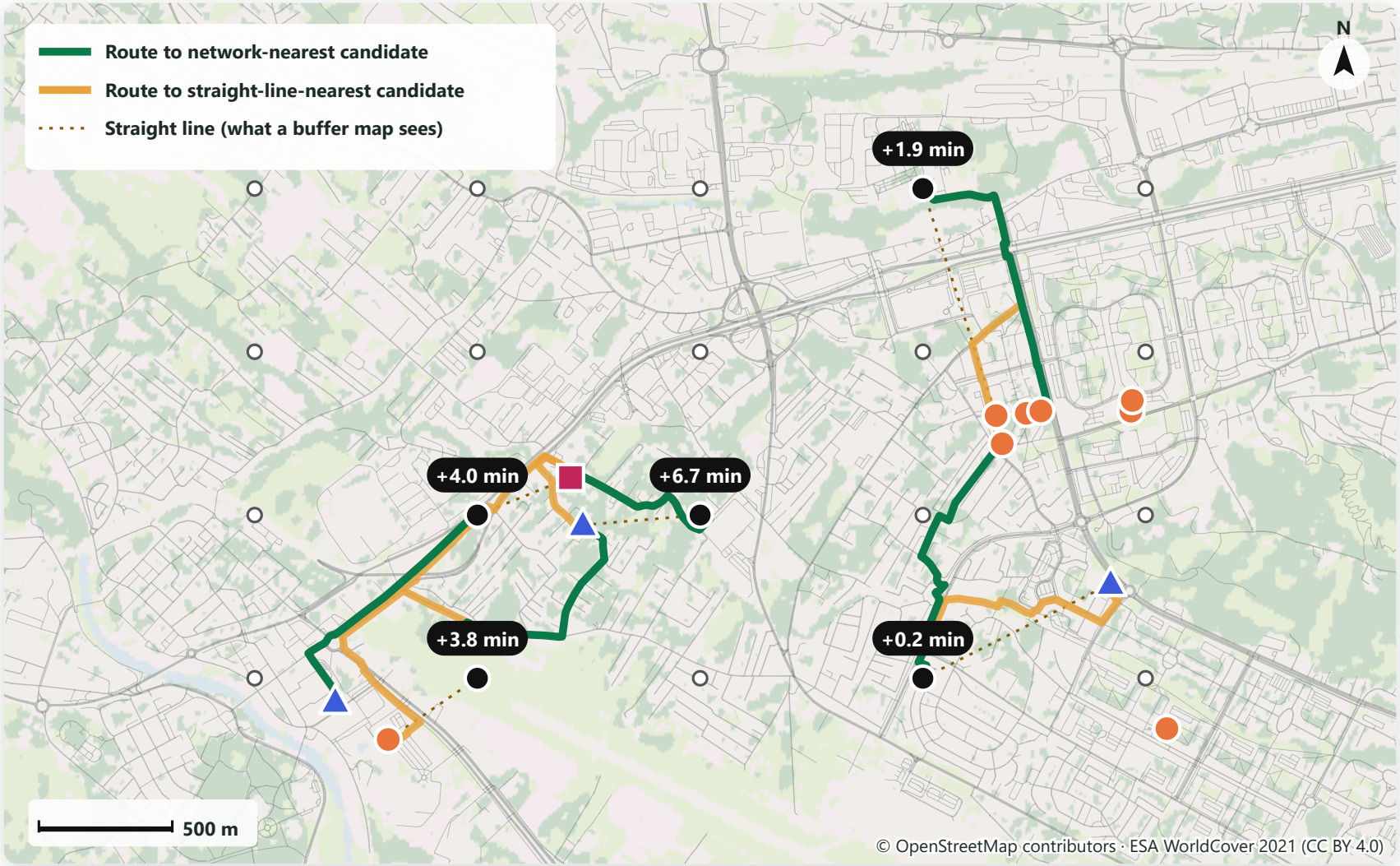
Equal destination weights. Thresholds are analytical scenarios, not health or planning standards.

The screening scenario in space



Playground candidates within 20 assumed walking minutes with ≥20 % vegetation in 300 m. Counts are candidate records at synthetic origins, not catchments or population coverage.

Finding 1 · Network proximity changes the nearest destination



Five origins where the straight-line-nearest candidate (any category) is not the network-nearest. Labels: extra walking minutes.

4 / 19
origins: nearest playground changes

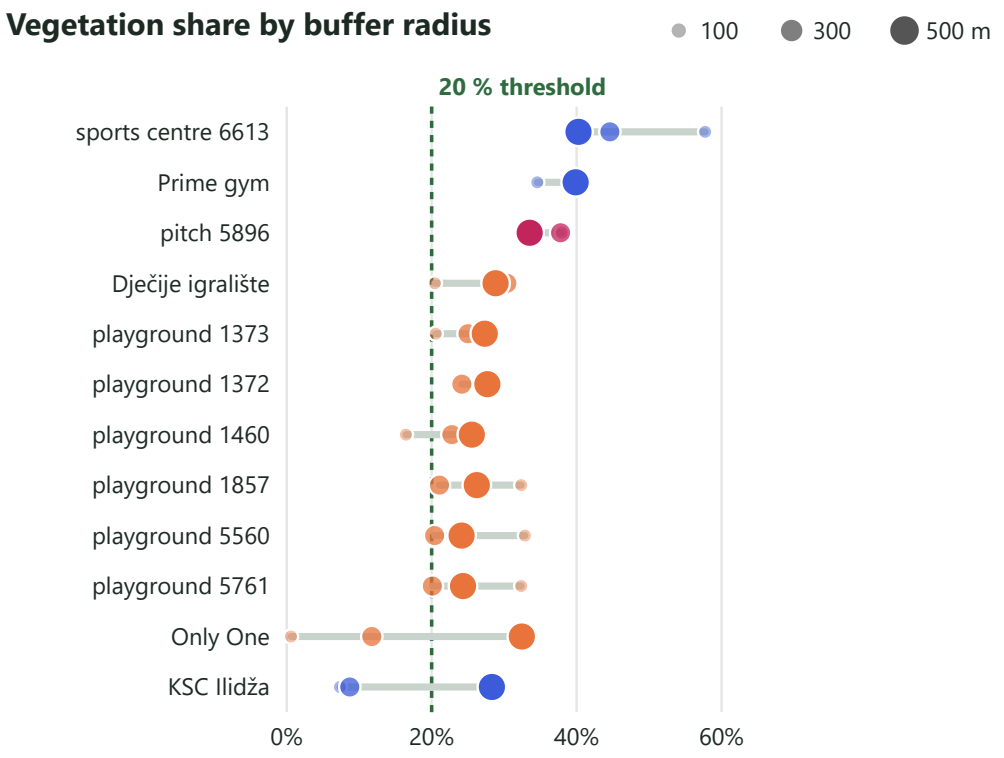
3 / 19
origins: nearest sports centre changes

9.98 min
largest extra walk (sports centre)

Single pitch: 0 / 19, no alternative. Conditional on directionality, snapping and 1.2 m/s; describes synthetic origins, not residents.

Finding 2 · The scale of green context changes the compromise set





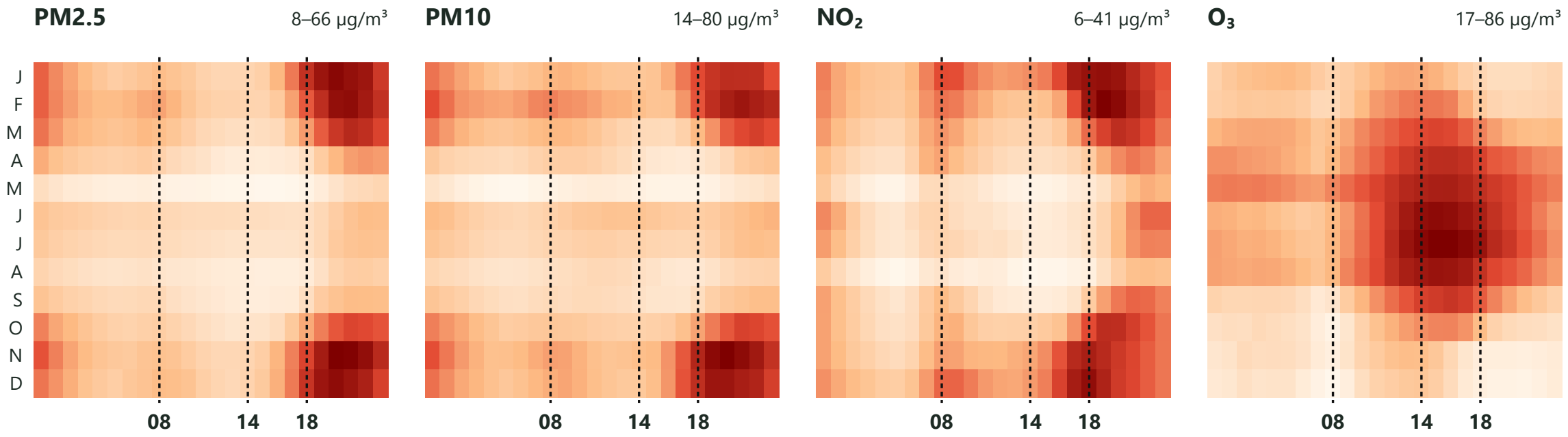
20 / 57
playground Pareto sets change, 100 → 500 m

0 / 57
sports centres and pitch

19 / 57
after rounding vegetation to 1 pp

81
of 1,620 rows unroutable, kept unknown

Finding 3 · Departure-time comparisons differ between pollutants



Mean hourly concentration by month (rows) and local hour, Europe/Sarajevo (columns). CAMS Europe 2021, one model cell. Dashed: compared departures 08 · 14 · 18.

−401
PM2.5 · median $E_{14}-E_{08}$ · 14:00 lower in 72.9 %

−225
PM10 · 14:00 lower in 44.8 %

−440
NO_2 · 14:00 lower in 78.4 %

+2,390
O_3 · 14:00 lower in only 0.6 %

µg·min/m³, 83,220 paired date–route scenarios per pollutant, 30-min stay. One regional 2021 series: retrospective, not a recommendation for today or a cleaner street.

Finding 4 · The representation itself is part of the result

Directionality

Removing graph direction rules shortens **125 / 228** routed pairs, max **11.45 min**. Sports-centre disagreement drops 3/19 → 2/19; playground set unchanged.

Candidate identity

Grouping the two unresolved close playground pairs changes opportunity counts in **15 / 57** origin–threshold cases (by at most two).

Connectors

Origin + destination connectors: median **2.13 %**, max **13.94 %** of baseline distance. All 20 origins satisfy 350 m snapping; 17 pass at 50 m, 19 at 100 m.

Independent rerouting reproduces all baseline distances. A record count should not be presented as an inventory of independent venues.

Controls and reproducibility

4,380 nested-interval monotonicity checks: no violations

80 origin–activity records and **1,620** rows verified; independent PM2.5 integration

10 core outputs regenerated twice offline, SHA-256 identical

Node.js standard library only, no npm dependencies

Station audit: official Ilidža displays exist, no matched authorised archive → no validation claim

Versioned analytical supplement with scripts, outputs, figures and manifests

Computational reproducibility, interpretive plausibility and real-world validity are three separate things. Tests and hashes establish only the first.

From indicator to planning question

Indicator	Concrete question	Permissible preliminary use	Does not establish
Network vs. straight-line; directional & connector sensitivity	Is the detour due to the city or the representation?	Review connections, crossings, entrances in records	A missing crossing or a justified construction project
Opportunity counts; unresolved grouping	How many distinct candidates remain?	Shortlist for documentary review (incl. physical education)	Capacity, suitability, permission, demand
Vegetation share; radius-dependent Pareto	Immediate setting or wider neighbourhood?	Examine spatial context of alternatives	Shade, usable green area, design quality
Pollutant-specific integrals; meteorology	Does the timing comparison change between pollutants?	Argue for temporal, pollutant-specific context	A safe time today or a cleaner route
Access documentation gaps	Which missing fact could invalidate the shortlist?	Assign a focused verification task	Access merely because an object is mapped

Example: the 9.98-minute sports-centre detour first suggests inspecting a missing link. The directionality audit shows the comparison changes with graph assumptions, so the first action is to inspect the representation, not to propose a physical connection.

Sport, physical education and health

Urban planning

Distribution and connection of potential activity settings

Urban design

Quality and continuity of route and place

Sport & PE practice

Can a specific activity be organised here for a specific group?

SportSpace supplies preliminary spatial information to this shared problem: which candidates to check, which cost more travel in the model, which details matter before organising activity.

Equipment, group size, supervision, surface, toilets, opening hours and permissions remain outside the analysed data. Health is upstream and conditional: opportunity ≠ participation ≠ benefit. The model cannot balance exercise benefit against pollution risk.

Limits and what comes next

Limits

- Synthetic origins: no population, schools, deprivation or demand
- 2026 network vs. 2021 environment: a mixed-year scenario
- Regional air and meteorology cannot differentiate nearby paths
- Pedestrian semantics, topology and connectors not field-verified
- Candidate identities and access incompletely verified
- No participants, activity minutes or health outcomes observed

Next, within the existing-data boundary

- Check pedestrian semantics and graph topology against source records
- Resolve close candidate identities; obtain documented venue properties
- Match authorised station archives if available
- Independently designed evaluation of decision usefulness

No CFD, new field campaigns or clinical research required for this contribution.

Conclusion

Network representation, the scale of vegetation context and the time of an assumed outing can each alter comparisons among mapped recreation candidates. Directionality and unresolved candidate identity do too.

The method organises the examination of **access, setting and timing** for sport and recreation, including the initial review of candidate settings for physical education. It cannot establish venue suitability, participation, equitable provision or health effects.

Defensible contribution: making the link from an indicator to a planning question and the next necessary evidence explicit, before a map-based result becomes a design or programme decision.

THANK YOU

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Live pilot, paper and supplement (BS / EN / TR):
3esign.github.io/sportspace-pilot

Data: OpenStreetMap © contributors · ESA WorldCover 2021 (modified Copernicus Sentinel data) · CAMS and ERA5 via Open-Meteo · FHMZ BiH